

K-Means Clustering Algorithm

K-Means Clustering is an Unsupervised Learning algorithm, which groups the unlabeled dataset into different clusters. Here K defines the number of pre-defined clusters that need to be created in the process, as if $K=2$, there will be two clusters, and for $K=3$, there will be three clusters, and so on.

It is an iterative algorithm that divides the unlabeled dataset into k different clusters in such a way that each dataset belongs only one group that has similar properties.

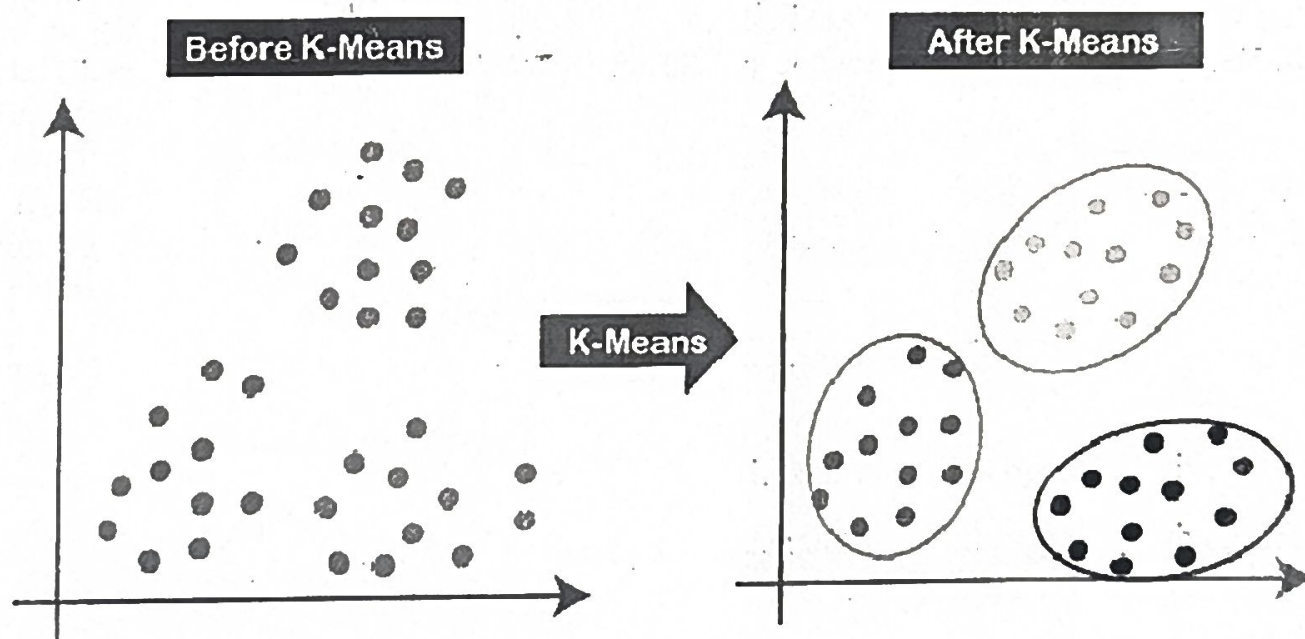
It allows us to cluster the data into different groups and a convenient way to discover the categories of groups in the unlabeled dataset on its own without the need for any training.

It is a centroid-based algorithm, where each cluster is associated with a centroid. The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.

The algorithm takes the unlabeled dataset as input, divides the dataset into k -number of clusters, and repeats the process until it does not find the best clusters. The value of k should be predetermined in this algorithm.

The k -means clustering algorithm mainly performs two tasks:

- Determines the best value for K center points or centroids by an iterative process.
- Assigns each data point to its closest k -center. Those data points which are near to the particular k -center, create a cluster.



How does the K-Means Algorithm Work?

Algorithm: k -means algorithm for partitioning, where each cluster's centre is represented by the mean value of the object in the cluster:

Input:

- k : the number of clusters.
- D : a data set containing n objects.

Output: A set of k clusters

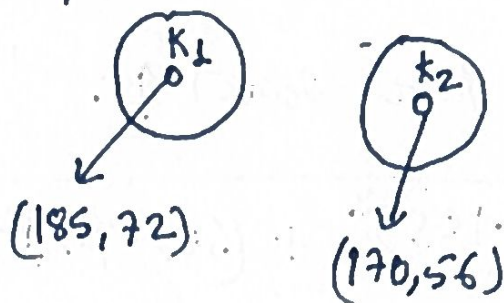
Method:

- (1) arbitrarily choose k objects from D as the initial cluster centers
- (2) repeat
- (3) (re)assign each object to the cluster to which the object is the most similar, based on the mean value of the objects in the cluster.
- (4) update the cluster means, that is, calculate the mean value of the objects for each cluster.
- (5) until no change.

K-mean clustering example:

S.No	Height	Weight
1	185	72
2	170	56
3	168	60
4	179	68
5	182	72
6	188	77
7	180	71
8	180	70
9	183	84
10	180	88
11	180	88.67
12	177	76

Step 1:



Step 2: Take 3 row for clustering. It is the euclidean distance based. That is as below

$$\sqrt{(x_0 - x_c)^2 + (y_0 - y_c)^2}$$

x_0 - observed value
 x_c - centroid value

here row 3 is observed value.
 k_1 & k_2 are centroid value

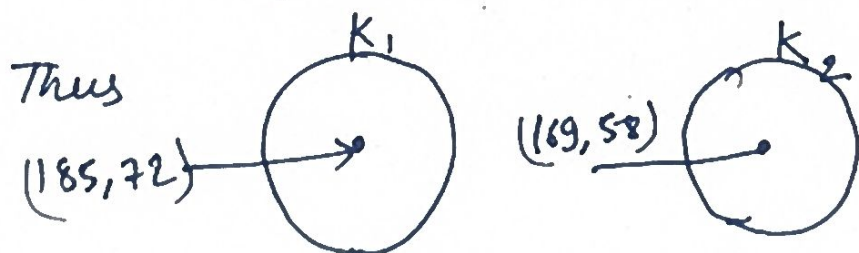
$$k_1 = \sqrt{(168 - 185)^2 + (60 - 72)^2} = 20.80$$

$$k_2 = \sqrt{(168 - 170)^2 + (60 - 56)^2} = 4.48$$

Accordinging k_1 & k_2 value row 3 data point belongs to k_2 . Therefore we must compute again centroid for k_2 .

Step 3: New centroid computation for k_2

$$k_2 = \left(\frac{170 + 168}{2}, \frac{60 + 56}{2} \right) = (169, 58)$$



$$K_1 = \{1\}$$

$$K_2 = \{2, 3\}$$

Now consider for row 4.

$$K_1 = \sqrt{(179 - 125)^2 + (68 - 72)^2} = 6.32$$

$$K_2 = \sqrt{(179 - 169)^2 + (68 - 58)^2} = 14.14$$

$$\therefore K_1 = \{1, 4\}$$

$$K_2 = \{2, 3\}$$

Similarly compute for other rows data
 point finally we get following cluster

$$K_1 = \{1, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$K_2 = \{2, 3\}$$

The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

Step-4: Calculate the variance and place a new centroid of each cluster.

Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.

Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.

Step-7: The model is ready.